

SURFACE OZONE VARIABILITY AND METEOROLOGICAL DRIVERS AT COASTAL AND MOUNTAIN SITES IN BULGARIA

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Abstract

The study is based on simultaneous measurements of surface ozone concentrations and meteorological parameters over a period of about two years in mountainous and coastal regions in Bulgaria. The average O_3 at the mountain station (Rozhen) is 48 ppb, and at the coastal station (Ahtopol), 41 ppb. It is found that extremely low and high ozone concentrations are less common in the mountainous area than in the coastal area. The diurnal cycle of O_3 at Rozhen shows small amplitude, with maximum values recorded at night, while at Ahtopol the diurnal cycle is distinct, with a maximum in the late afternoon and a minimum in the morning. The seasonal variation in ozone is well defined at both locations. There is a broader spring–summer maximum at the mountain station, while at the coastal station, the maximum is limited to the summer. Nonparametric regression analysis showed that ozone levels at Rozhen are independent of wind, while sea winds bring ozone-rich air to Ahtopol. Boosted regression tree modelling explained 87% and 81% of O_3 variability at the mountain and coastal sites, respectively, reproducing expected meteorological influences while capturing site-specific effects and variable interactions.

Key words: surface ozone, meteorological drivers, regression analysis, diurnal and seasonal variability, mountain and coastal areas

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Introduction. Ozone (O_3) is a trace gas with contrasting roles in the atmosphere. In the stratosphere, it protects life from harmful ultraviolet radiation [1], while in the troposphere it contributes to radiative forcing as a greenhouse gas [2] and acts as an air pollutant due to its toxicity. Surface ozone is a secondary pollutant formed through photochemical reactions involving volatile organic compounds (VOC) and nitrogen oxides (NO_x), with the formation regime depending on the VOC/ NO_x ratio [3]. O_3 concentrations are influenced not only by precursor emissions but also by meteorological conditions, deposition, long-range transport, and stratosphere–troposphere exchange, which together drive O_3 seasonal and diurnal variability [4]. The spring–summer ozone maximum observed across the Northern Hemisphere is generally attributed to a combination of stratospheric transport and photochemical production [5]. In rural areas, ozone typically peaks during the afternoon and reaches a minimum at night due to variations in boundary-layer mixing and deposition. Topography further modulates these nocturnal changes – O_3 generally declines more strongly in low-lying areas, whereas elevated sites and coastal regions tend to maintain higher nighttime O_3 due to enhanced turbulence and advection [6]. Meteorological factors such as temperature, solar radiation, humidity, and wind strongly affect ozone concentrations, and their relative importance varies by environment [7]. Coastal sites are mainly influenced by local meteorology and air mass transport [8], whereas high-altitude stations are more strongly affected by atmospheric dynamics, including synoptic-scale circulation patterns and long-range transport, with photochemical production playing a secondary role [9]. Since meteorological conditions exert a strong control on ozone variability, they should be accounted for when analysing and comparing O_3 concentrations across different locations [10].

This study compares the temporal behaviour of surface ozone at mountain and coastal sites in Bulgaria, focusing on diurnal and seasonal variability and meteorological influences. It applies nonparametric methods and boosted regression tree models to assess the relative importance of meteorological drivers and transport processes in shaping O_3 concentrations at both locations.

Data and methods. For the purposes of this study, hourly data from 7 July 1999 to 11 September 2001 were analysed, including surface O_3 and key meteorological variables (temperature, humidity, wind speed and direction, solar radiation, precipitation, and atmospheric pressure). Measurements were obtained at two fixed sites: a coastal station (Ahtopol, $42.08^\circ N$, $27.95^\circ E$, 20 m a.s.l.) situated near a mixed forest, influenced by marine conditions and potential local anthropogenic activity, and a mountain station (Rozhen, $41.69^\circ N$, $24.74^\circ E$, 1750 m a.s.l.), located in a remote forested area of the Rhodope Mountains (Fig. 1). This dataset is likely to provide one of the first simultaneous, high-resolution (hourly) multi-year ozone and meteorological records at rural sites in Bulgaria. Further details on the stations and instrumentation are provided in [11].

During data preparation, several gaps were identified. Atmospheric pressure

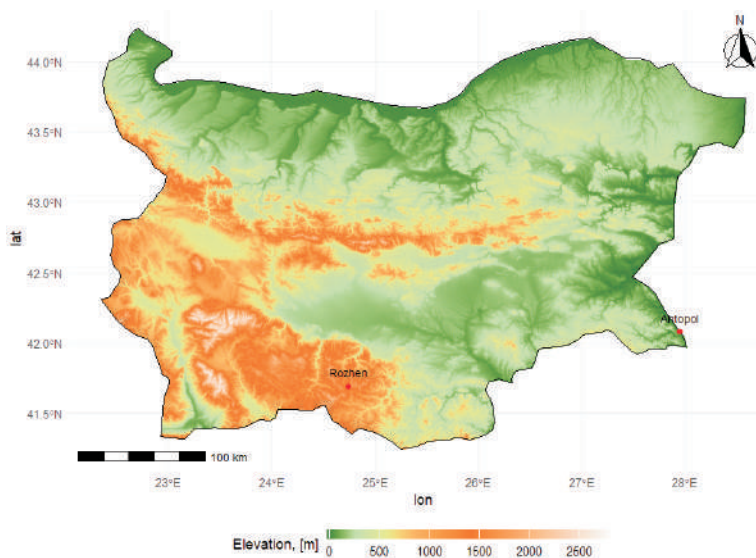


Fig. 1. Locations of the Rozhen and Ahtopol sites in Bulgaria

was largely missing at Rozhen (67.5%) and absent at Ahtopol, while smaller gaps occurred for O_3 (5.8% at Rozhen) and for wind direction (5.7% at Ahtopol). Pressure data were reconstructed using ERA5 reanalysis [12] (Ahtopol) and a simple linear model based on ERA5 pressure (Rozhen). Remaining missing values were imputed using a random forest approach [13] with available measurements and ERA5 variables as predictors.

A comprehensive statistical package [14] was applied to analyse O_3 data and the role of meteorological conditions at the two locations. Monthly and diurnal variations in mean O_3 concentrations were computed for each station and visualised using heatmaps.

To assess the influence of local transport and potential O_3 sources, a non-parametric wind regression approach [15] was applied. The results are presented as bivariate polar plots of O_3 concentration as a function of wind speed and wind direction.

Since surface O_3 depends on transport, photochemical production, and deposition – all influenced by meteorology – meteorological normalisation was applied [16]. For this purpose, a boosted regression trees model was used to predict O_3 from temporal variables (hour of day, day of year) and meteorological predictors (temperature, vapour pressure deficit, solar radiation, atmospheric pressure, wind speed, and wind direction). Hyperparameters of the model were optimized using 10-fold cross-validation to minimise the loss function (sum of squared errors). Relationships between O_3 and individual predictors were examined using partial dependence plots, which show the marginal effect of a given variable after accounting for the average effects of all other variables in the model. These plots also

display variable importance based on split frequency weighted by improvements in model fit.

Results. Over the study period, surface ozone at Rozhen and Ahtopol showed means of 48 and 41 ppb, medians of 48 and 41 ppb, and maxima of 103 and 102 ppb, respectively. O_3 distributions were approximately symmetric at both sites, with slight right skewness at Ahtopol (0.28) and near-zero skewness at Rozhen. Excess kurtosis values of -0.43 (Rozhen) and 0.23 (Ahtopol) indicate a platykurtic distribution at Rozhen and a slightly leptokurtic one at Ahtopol, suggesting fewer extremes at Rozhen and a somewhat higher likelihood of unusually high or low ozone values at Ahtopol.

Figure 2 illustrates the monthly and diurnal variation of O_3 at both stations. A clear seasonal cycle is observed, with lower concentrations in the cold season and a pronounced summer maximum. At Rozhen, elevated levels persist from April to August (50–60 ppb), whereas at Ahtopol the peak is confined to July–August, reaching ≈ 60 ppb in the afternoon. Both sites show a minimum in December (≈ 35 ppb at Rozhen and ≈ 30 ppb at Ahtopol). Despite site differences, the similar seasonal patterns likely reflect large-scale atmospheric controls on O_3 variability.

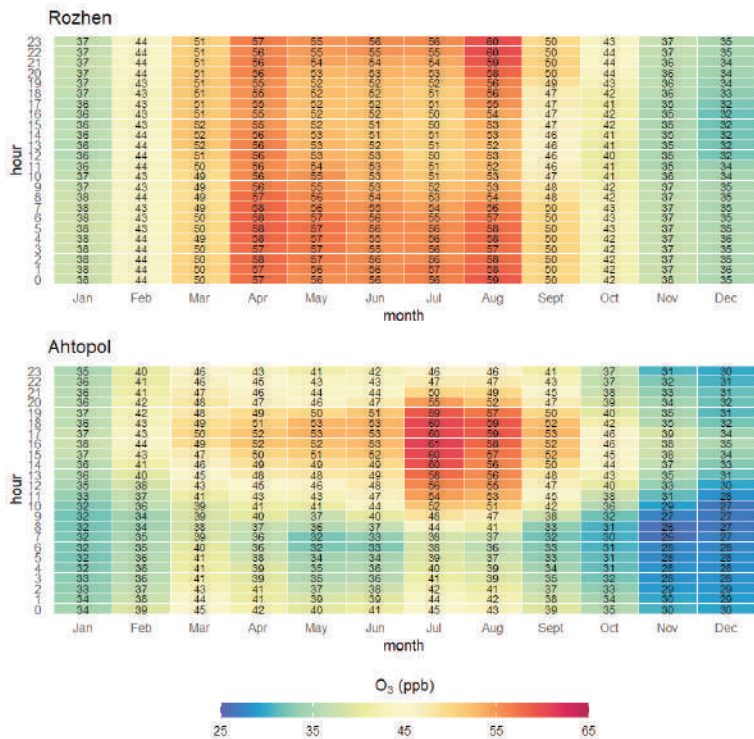


Fig. 2. Monthly–hourly surface O_3 at Rozhen (top) and Ahtopol (bottom). Colours indicate mean ozone concentrations, highlighting the combined diurnal and seasonal structure at each site

Figure 2 also highlights distinct contrasts in diurnal ozone behaviour between the two stations. At Rozhen, the diurnal cycle is weak, with amplitudes of 1–2 ppb in winter and up to ~ 4 ppb in August. Maximum concentrations occur at night and minimum concentrations in the afternoon, likely related to widespread nocturnal ozone enhancement (NOE) observed in multiple regions [17], additionally amplified by downward transport and mixing of ozone-rich air from the free troposphere or nighttime residual layer at elevated sites. At Ahtopol, the diurnal cycle is much stronger, especially in summer, reaching 11–12 ppb amplitude in August, but reduced to 3–6 ppb in winter. This pattern suggests a dominant role of photochemical production at the coastal site, although thermally driven local circulations may also contribute.

The polar plots in Fig. 3 do not show consistent relationship between wind direction and surface ozone at Rozhen, suggesting the absence of dominant local sources and indicating that the site reflects regional background conditions. In contrast, the highest O_3 concentrations at Ahtopol occur under southeasterly winds, particularly in summer, while lower levels are associated with westerly flow. Elevated ozone during easterly winds likely reflects transport of ozone-rich

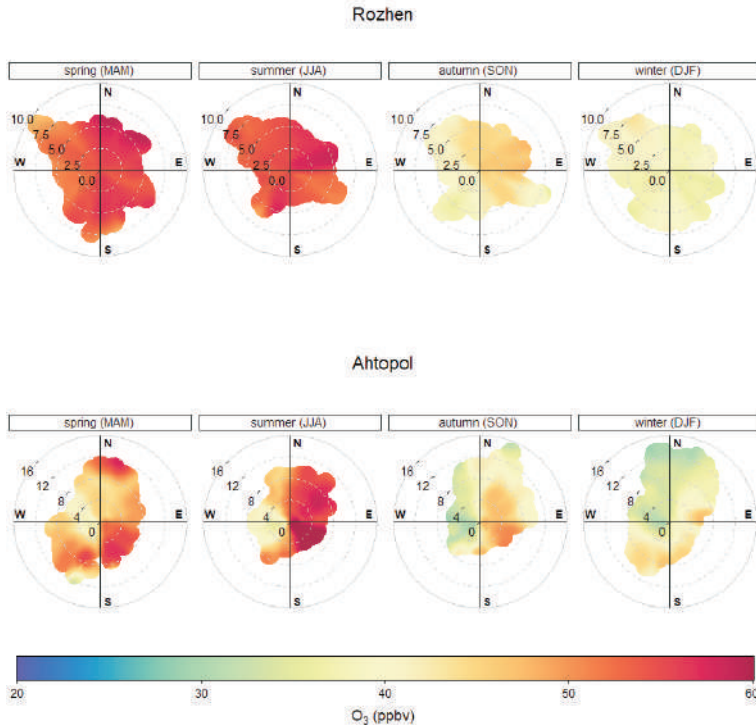


Fig. 3. Bivariate polar plots of O_3 as a function of wind speed and direction, arranged by station (Rozhen, top; Ahtopol, bottom) and by season (spring to winter)

marine air masses. At both sites, O_3 concentrations are generally higher in spring and summer and lower in autumn and winter.

Modelling surface O_3 using boosted regression trees with the same covariates (both time-related and meteorological) at Rozen and Ahtopol yielded good predictive performance, with the models explaining 87% and 83% of the variance in O_3 concentrations, respectively. However, as can be seen in Fig. 4, the relative importance of the predictors differs substantially between the two stations. At Rozen, the day of the year (yday) is the most influential predictor (41.2%), followed by air temperature (27.9%), atmospheric pressure (11.6%), vapour pressure deficit (VPD) (8.9%), and the remaining variables. At Ahtopol, VPD is the dominant predictor (47.8%), followed by yday (18.1%), air temperature (13.1%), wind direction (7.3%), and the other covariates.

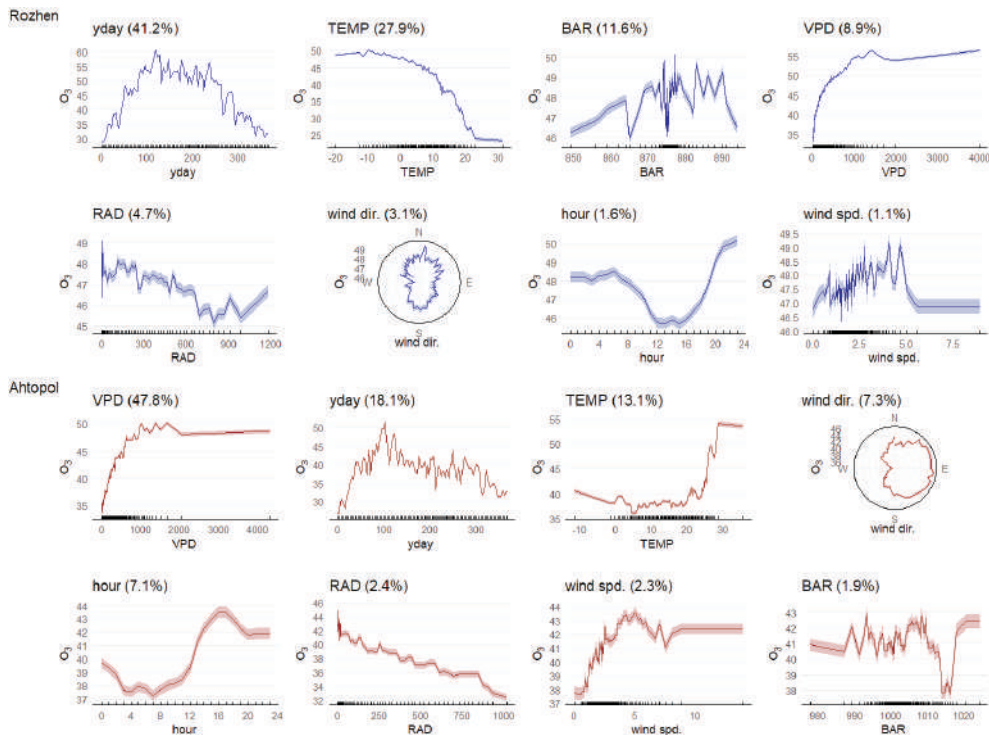


Fig. 4. Partial dependence plots showing the response of surface ozone at Rozen (upper, blue) and Ahtopol (lower, red) to temporal and meteorological predictors in the boosted regression tree model. Curves represent marginal predictor effects. Shaded bands indicate bootstrapped 95% confidence intervals. Relative importance is given in parentheses

The partial dependence plots for day of the year and hour of the day (Fig. 4) confirm the previously identified seasonal and diurnal ozone patterns. The relationships with meteorological variables are particularly notable. At Rozen, ozone decreases with increasing temperature, whereas at Ahtopol it increases. The

effect of VPD is similar at both sites, although its importance differs, with higher VPD associated with elevated O_3 concentrations, possibly due to reduced stomatal uptake during drier conditions. Figure 4 also confirms higher O_3 at Ahtopol under easterly (marine) winds. In contrast, wind direction is less important at Rozhen and exerts a weaker influence on ozone. Solar radiation makes a relatively small contribution to model performance at both sites; the apparent decrease in O_3 concentrations with increasing radiation may reflect interactions among predictors.

Conclusions. The analysis of surface O_3 at two sites with contrasting geographical characteristics highlights the influence of local conditions and meteorological drivers on O_3 variability. Mean concentrations during the study period were 48 ppb at Rozhen and 41 ppb at Ahtopol, consistent with previous findings [11]. Extreme O_3 values occurred more frequently at Ahtopol. The two stations exhibited contrasting diurnal patterns: Ahtopol showed the typical morning minimum and afternoon maximum, whereas Rozhen displayed a daytime minimum and nighttime maximum, with weaker diurnal variability overall. Wind analysis indicated no clear directional dependence at Rozhen, supporting its representativeness of regional background conditions, while elevated O_3 at Ahtopol was associated with ozone rich marine air masses transported onshore. Boosted regression tree models showed strong performance, explaining 87% and 83% of O_3 variance at Rozhen and Ahtopol, respectively. The identified relationships between O_3 and meteorological and temporal predictors were consistent with expected physical processes, while also reflecting predictor interactions. Future studies could further assess ozone exposure and risk metrics relevant to human health and vegetation.

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